

Religious Texts Platform

Technical Documentation

Version 0.1 | May 2026

1. Project Overview

The Religious Texts Platform is a Vaadin-based web application for reading, comparing, and studying religious texts side-by-side. The system ingests texts from multiple sources into a BaseX XML database and presents them in a multi-column reader interface.

The platform is designed to support Bible translations, the Quran, Hadith collections, and associated commentaries in a single unified reader.

1.1 Technology Stack

Layer	Technology
Frontend	Vaadin 24 (Java server-side rendering)
Backend	Spring Boot 3.2
Database	BaseX 10 (native XML database)
Build	Maven (multi-module)
Data ingestion	Spring Boot service on port 8091
App server	Embedded Tomcat on port 8090
Bible source	Local git clone (w/deh/bible-api) + API.Bible

1.2 Module Structure

- `app/` — Vaadin frontend + Spring Boot app (port 8090)
- `ingestion/` — Ingestion service (port 8091)
- `schema/` — XSD schema for the `religioustext` XML namespace
- `docker/` — Docker configuration for BaseX

2. Data Architecture

2.1 XML Schema

All texts are stored in BaseX under the namespace `http://religioustext.org/schema/1.0`. The root element `<text>` carries translation-level metadata; books, chapters, and verses are nested within.

```
<text id="bible-kjv-1611" type="bible" translation="King James Version"
      abbreviation="KJV" bcp47Language="en" direction="ltr"
      license="Public Domain" year="1611" totalVerses="36820">
  <book name="Genesis" code="GEN" canonicalOrder="1" testament="OT">
    <chapter number="1" title="The Creation">
      <verse number="1" bookName="Genesis" chapterNumber="1">
        In the beginning God created...
      </verse>
    </chapter>
  </book>
</text>
```

2.2 Key Attributes

Element	Attribute	Purpose
<code><text></code>	<code>id</code>	Document ID e.g. bible-kjv-1611
<code><text></code>	<code>totalVerses</code>	Pre-computed verse count
<code><book></code>	<code>code</code>	Canonical API.Bible code (GEN, EXO... REV)
<code><book></code>	<code>canonicalOrder</code>	Sort order (1=Genesis, 66=Revelation)
<code><book></code>	<code>testament</code>	OT / NT / DC (Deuterocanonical)
<code><chapter></code>	<code>title</code>	Editorial section title (where available)
<code><verse></code>	<code>number</code>	Verse number within chapter

2.3 Translation-Independent Verse Anchors

Each verse div in the UI is assigned an HTML anchor ID in the format `verse-{CODE}-{chapter}-{verse}`, for example `verse-GEN-1-1`. This is translation-independent — the same anchor ID is used across all open Bible columns, enabling commentary cross-references to highlight the same passage in every open translation simultaneously.

3. Ingestion Pipeline

3.1 Architecture

The ingestion service uses a producer/consumer pipeline with a configurable thread pool (default 5 crawler threads) feeding a single insert thread via a BlockingQueue. A poison pill sentinel terminates the insert thread cleanly after all books are processed.

3.2 Local Clone Ingestion (LocalIngestionService)

Reads Bible text from a local clone of the wldch/bible-api repository. Chapter titles are optionally fetched from API.Bible (250ms rate limit between calls). Book metadata (canonical order, API code, testament) is resolved via a BOOK_META map keyed on folder name.

- Source path: /Volumes/VMs/data/bible-api-source/bibles/
- Supports English, Spanish, German, Greek, and other language folders
- BOOK_META covers Protestant canon + full DC/Apocrypha + Spanish (RVR09) folder names

3.3 API.Bible Ingestion (IngestionService)

Used for licensed translations (NIV, NASB20, NBLA) where the full text must be fetched from the API. Rate-limited; uses 3 licensed API.Bible slots.

3.4 Current Ingestion Status

Translation	Language	Verses	Status
NIV 2011	English	30,752	✓ Complete — abbreviated book names, no code attr yet
KJV 1611	English	36,820	✓ Complete — inc. Apocrypha (80 books)
ASV 1901	English	31,102	✓ Complete
WEB	English	~31K+	✓ Complete — inc. extra DC books
DRA 1899	English Catholic	~46K	✓ Complete
RVR09 1909	Spanish	31,102	✓ Complete
NASB20	English	—	⌚ Not yet ingested
NBLA	Spanish	—	⌚ Not yet ingested

3.5 Known Issues

- NIV book names are abbreviated (Gen., Lev. etc.) — needs re-ingestion via API.Bible path to fix
- NIV <book> elements have no code= attribute — patched in all other translations via XQuery update
- Document naming inconsistency: NIV stored as bible-niv-2011.xml (with .xml suffix), others without

4. Application Layer

4.1 TextQueryService

Queries BaseX via its REST API using XQuery over HTTP. The `executeQuery()` method uses `UriComponentsBuilder` to avoid double URL-encoding of query parameters. All verse data is returned as pipe-delimited rows and parsed into `VerseRef` objects.

4.2 VerseRef Model

Immutable record carrying: `sourceId`, `bookName`, `bookCode`, `chapterNumber`, `chapterTitle`, `verseNumber`, `content`, `globalCanonicalSeq`, `globalChronologicalSeq`, `globalNarrativeSeq`, `note`.

4.3 DisplayOptions

Mode	Description
Original	Scriptio continua — all caps, no spaces or punctuation, continuous block
Chapters	Chapter headings only, prose verse flow
Verses (default)	Chapter headings + superscript verse numbers
Titles	Chapters + verse numbers + editorial section titles

4.4 ReaderView (UI)

Multi-column Vaadin layout. Each column is 33vw wide, horizontally scrollable. Columns are independent but sync by default — changing book/chapter in one synced column updates all other synced columns. Sync can be broken per column via the chain-link icon.

- Source selector: per-column ComboBox populated from `BaseX listSources()`
- Book/chapter nav: Select dropdowns with prev/next chapter buttons
- Sync toggle: LINK / UNLINK icon per column
- Display mode: per-column dropdown (Original / Chapters / Verses / Titles)

5. Notable Bug Fixes

Bug	Fix
canonicalOrder was alphabetical (folder discovery order)	Use BOOK_META.canonicalOrder() not thread submission counter
InterruptedException noise on insert thread	Replaced interrupt() with poison pill (sentinel CrawledBook)
BaseXStore proc.waitFor() InterruptedException	Replaced with waitFor(30, TimeUnit.SECONDS) + destroyForcibly()
BOOK_META folder name mismatches (ecclesiasticus, manasseh, bel, esther(greek))	Fixed keys to match actual folder names
RVR09 Spanish folder names not in BOOK_META	Added full Spanish mapping (génesis, éxodo... apocalipsis)
Double URL encoding in XQuery REST calls	Replaced URLEncoder + string URL with UriComponentsBuilder
book code= attribute missing from XML	Added to serialiseBook(); patched existing docs via XQuery update
Vaadin theme directory missing	Created app/src/main/frontend/themes/religious-texts/

6. Planned Features

6.1 Continuous Scroll

Infinite scroll with IntersectionObserver — loading next/previous chapters as the user scrolls. DOM window limited to previous + current + next book (~2,400 verse divs). Chapter selector updates as chapters cross viewport. Synced columns navigate (not pixel-scroll) to matching passage.

6.2 Quran & Hadith

Quran ingestion via fawazahmed0/quran-api. Hadith collections (Sahih al-Bukhari, Sahih Muslim, and the four Sunan) via a similar pipeline. Same XML schema with type="quran" and type="hadith".

6.3 Commentary & Cross-References

Commentary columns with clickable verse reference links. Links use the translation-independent anchor scheme (verse-GEN-1-1) to scroll all open Bible columns to the referenced passage simultaneously. If the referenced text is not open, offer to open a new column.

6.4 Bulk Translation Import

The local clone contains 200+ translations across dozens of languages. A future bulk import task will iterate all folders in /Volumes/VMs/data/bible-api-source/bibles/, auto-generate bible-

sources.yml entries, and ingest everything. Requires language detection from folder name prefix and smarter BOOK_META fallback for non-Latin scripts.

6.5 User Features

- Spring Security login
- Personal notes column
- Verse highlighting and bookmarks
- Search across all ingested texts